

Land Management

Final Presentation

Manikandan Somu

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Planning Task

- To develop a mixed-used, medium-density urban area.
- There should be shops or offices in the ground floor, offices in the first floor and flats in the upper floors.
- Traffic areas should be designed with all modes of transport (walking, cycling, public transport, cars)
- Provide for public amenities(green spaces, lot for a kindergarten, ...).
- Cater for the needs of a variety of dwellers(singles, couples, families, shared flats, old people's homes, ...)
- with different incomes.
- Your proposed development should fit into the surrounding.

Location:

Size : **about 5 hectares**

Location :
Endenich, Bonn, Germany



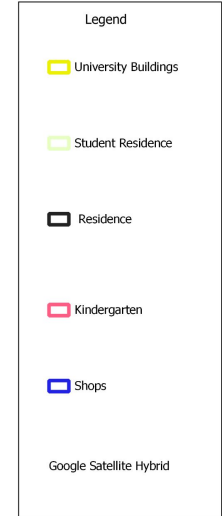
Site Appraisal

Parameters considered:

- LOCATION
- ACCESSIBILITY FOR TRANSPORTATION
- LAND USE STRUCTURE
- SENSORY COMPONENTS (traffic, sounds, smell etc..)
- TOPOGRAPHY
- MAN-MADE FEATURES

Land use structure:

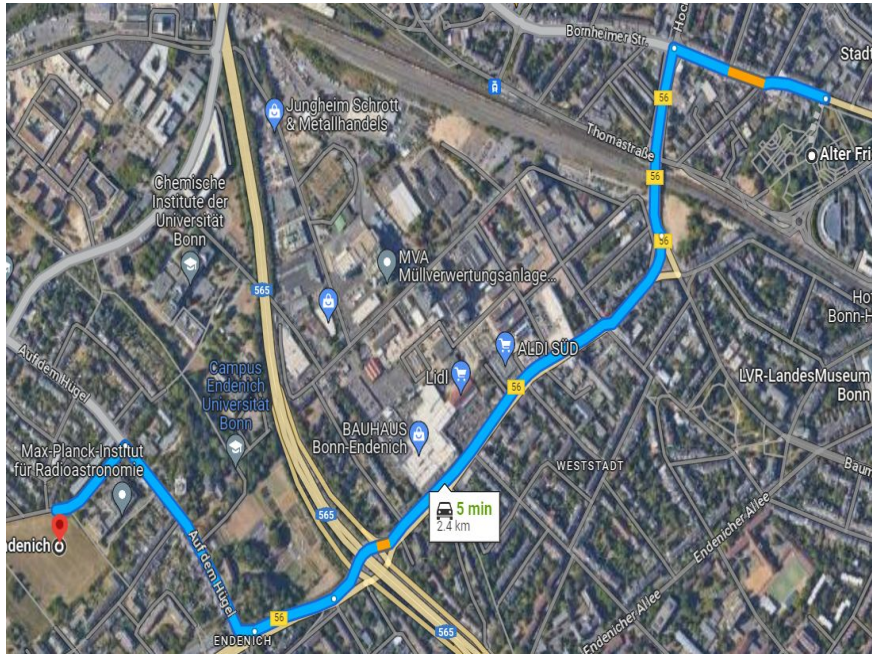
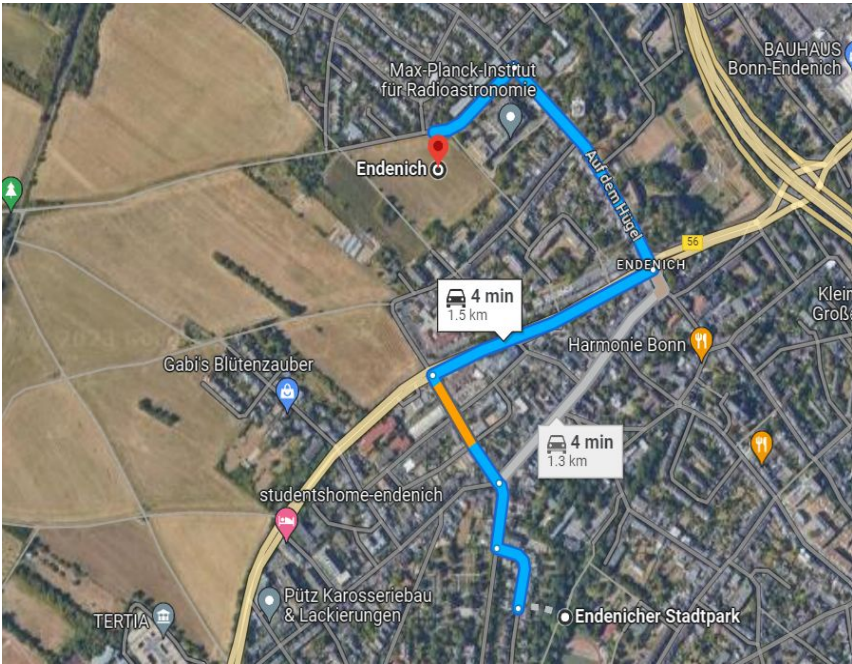
- Mixed Urban area and Medium Density Population.
- Mostly Residential, Some parts are Mixed urban, Single family houses, Student dorms, Apartments, Institutes.
- Commercial Shops are available like Petty Shops, Restaurants, ALDI, Kik etc
- Green spaces and Road Network are constructed well.
- Land Use Structure of the community is diverse and appears to be designed to support the needs of its residents



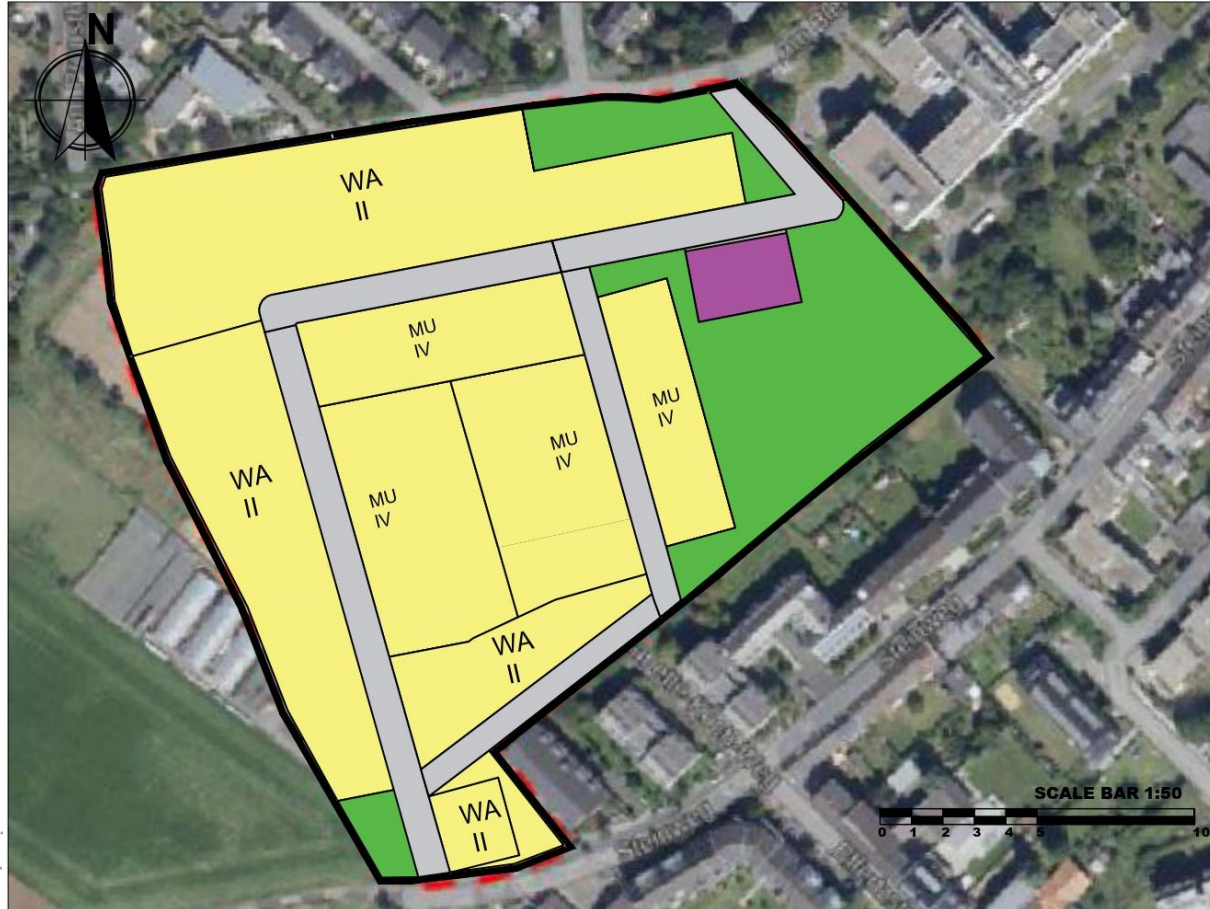
Natural conditions: (Topography, Geology, Water, Climate)

- Slope - Fairly Level
- Area - Suburb
- Elevation - 64 - 75 m.
- Soil - Laterite
- Pollution level - Low
- Wind direction - South West

Attraction:



STRUCTURAL CONCEPT PLAN



LEGEND

- WA Residential Area
- MU Mixed Used Area
- Kindergarten
- Green Areas
- Roads
- I-IV Number of Floors
- Boundary

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Juli, 2024

Structural plan

- No double connection
- Every house connected to the road
- Allotment of some space for public amenities
- With one kindergarten
- Roads of site connected to existing roads on either sides without disruption

Design Plan



LEGEND

- Public Green Areas
- Private Green Areas
- Pond
- Collecting Road
- Pedestrian/Bicycle Lane
- Kita
- PlayGround
- Footpath
- Garage
- Access
- Entrance
- I-IV Number of Floors
- DH Duplex House
- EH Single Unit House
- TH TownHouse
- Kindergarten Area
- Saddle Roof
- Flat Roof

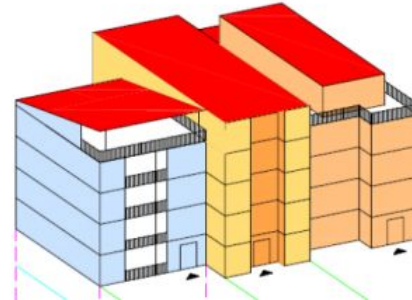
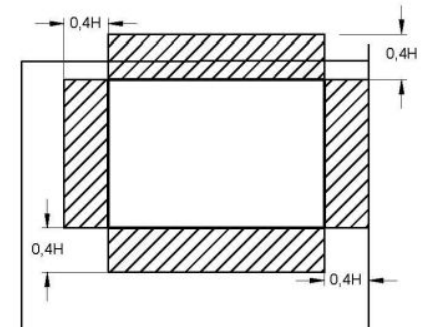
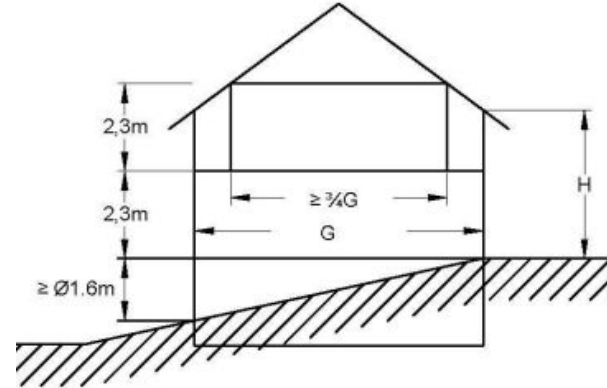
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Design plan

Construction methodology adopted to plan :

- Detached houses with saddle roof
- Single houses with saddle roof
- Town houses with flat roof, private spaces & underground garage



Design plan

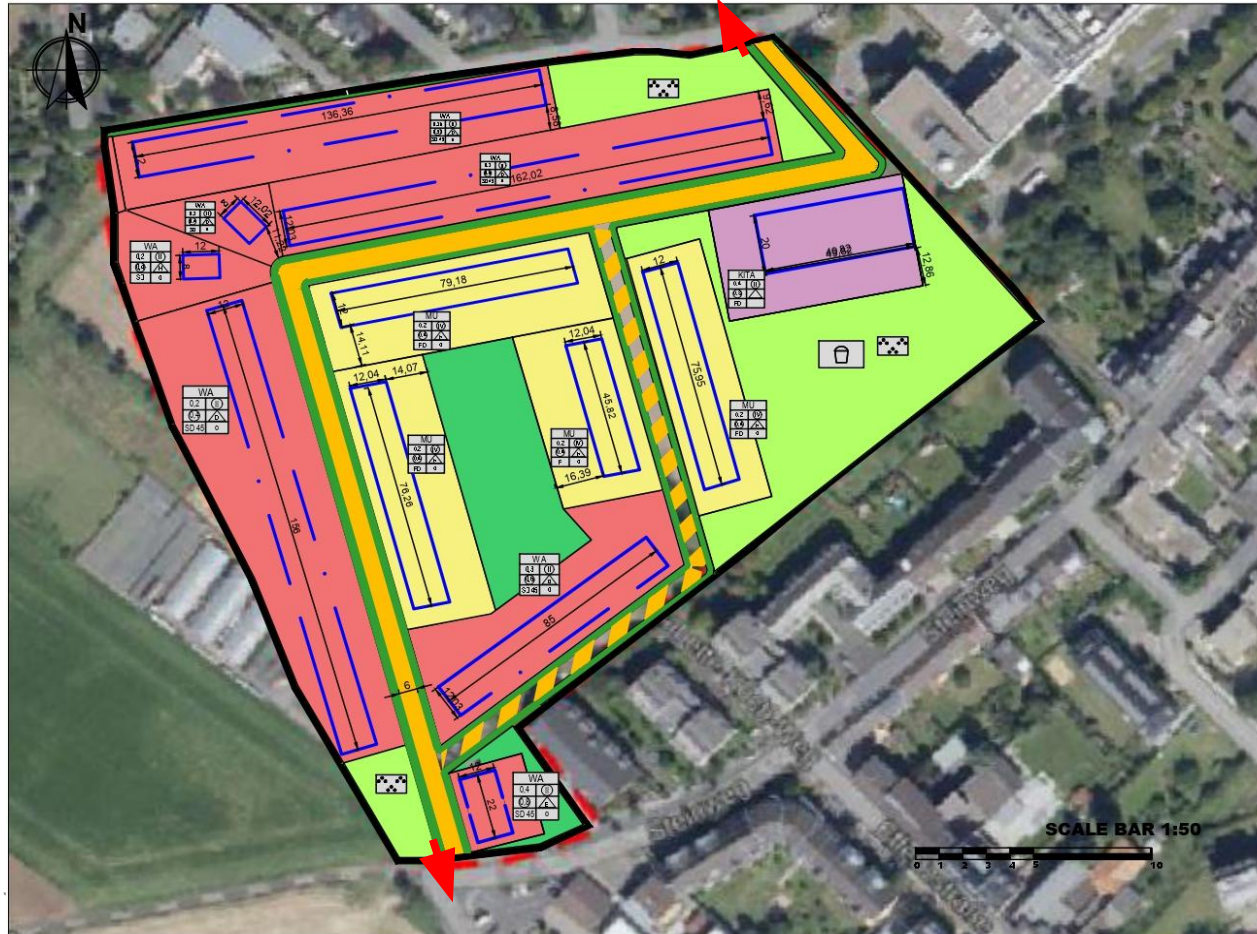
KinderGarten:

- It is walkable from residential place
- Attached to public green space
- With equipment of playgrounds has designed

Planned as per

Benchmarks for social infrastructure, Source: STÄDTEBAU-INSTITUT DER UNIVERSITÄT
STUTTGART 2001: 112

Binding LandUse Plan



LEGEND

- General Residential Area
- Mixed Used Area
- Kindergarten
- Public Green Areas
- Road Traffic Area
- Traffic Areas for Special Purpose
- Pedestrian/Bicycle Path
- Private Green Areas
- Building Boundary
- Area Boundary
- Park
- PlayGround
- WA/MU Building Area
- I - IV Number of Floors
- 0.4 Floor Space Index
- 0.2 Site Occupancy Index
- SD 45 Saddle Roof (Pitch)
- FD Flat Roof
- △ Structure

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Binding Land Use Plan

Urban development design in Germany is shaped by several building planning laws and regulations.

- 1. Federal Building Code (BauGB) §§ 1-33 is crucial for urban planning**
- 2. Federal Land Utilization Ordinance (BauNVO)**
- 3. Notation Symbol Ordinance (PlanZV)**
- 4. Building Regulations North Rhine-Westphalia (BauO NRW)**

Binding Land Use Plan

Urban Land Use Planning includes

1. Initial Draft and Early Participation:
 - Comments and suggestions from this stage can lead to adjustments in the plan
2. Second Participation Phase:
 - The public and interest groups can submit written comments and suggestions
3. Finalizing the Plan:
4. Discuss the finance, recompensation:
5. Adoption:

LAND READJUSTMENT PLAN

Original Layout (Before Land Readjustment)



Landowner:

- City of Bonn
- Owner A
- Owner B
- Owner C
- Owner D

Land value:

Land value before
planning: 275 €/m²

Land value after
planning: 440 €/m²

LAND READJUSTMENT PLAN



LEGEND

- Owner A
- Owner B
- Owner C
- Owner D
- City of Bonn
- Boundary

Land Value before Planning : €275m²

Land Value after Planning : €440m²

Endenich - Bonn

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	Area(m2)	Value	
Whole Area (U)	48,756		
Old Traffic Area(A)	5203		
Insertion Mass(E)	43,553	275	11977113.89
New traffic and development area(N)	16806.3743		
Area Deduction(f)	11603.3743		0.2664187686
Distribution Mass(V)	31,950	440	14057897.52
Distribution Quotient(q)			1.17372997

Owner A

		Value	
			€
Insertion Plot(E)	2127	275	584925
Target Claim(V_{i(a)})	1599.125845		703615.3716
Readjustment Benefit(G1)			118690.3716
Allocation Plot (Z_{i(a)})	1400	440	616000
Readjustment Benefit(G2)	-199.1258446		-87615.37162
Financial Compensation			31075

Owner B

		Value	
	m2		€
Insertion Plot(E)	6690	275	1839750
Target Claim(V_{i(b)})	5029.690597		2213063.863
Readjustment Benefit(G1)			373313.8628
Allocation Plot (Z_{i(b)})	4515.3076	440	1986735.344
Readjustment Benefit(G2)	-514.3829973		-226328.5188
Financial Compensation			146985.344

Owner C

		Value	
			€
Insertion Plot(E)	8079	275	2221725
Target Claim(V_{i(c)})	6073.97165		2672547.526
Readjustment Benefit(G1)			450822.5258
Allocation Plot (Z_{i(c)})	5349.5205	440	2353789.02
Readjustment Benefit(G2)	-724.4511495		-318758.5058
Financial Compensation			132064.02

Owner D

		Value	
	m2		€
Insertion Plot(E)	30232	275	8313800
Target Claim(V_{i(d)})	22729.08911		10000799.21
Readjustment Benefit(G1)			1686999.208
Allocation Plot (Z_{i(d)})	20,685	440	9101373.16
Readjustment Benefit(G2)	-2,044		-899426.0478
Financial Compensation			787573.16

Land Readjustment

Tried to retain the places of the initial owners

Allotted parcels at least close to the original parcels to please owners

References:

1. <https://www.google.com/maps>
2. LaGro, James A. Site Analysis: Informing Context-Sensitive and Sustainable Site Planning and Design. John Wiley & Sons, 2013.
3. https://stadtplan.bonn.de/cms/cms.pl?Amt=Stadtplan&set=0_0_0_0&act=0

Thank you 😊